

Practice Quiz 01C — MATH 2055 — Fall 2026 — Solutions

Practice for Quiz 01

Truth tables and logical equivalence.

1. Use a complete truth table to determine whether

$$\neg(P \wedge Q) \quad \text{and} \quad \neg P \wedge \neg Q$$

are logically equivalent. State your conclusion.

Solution

P	Q	$\neg(P \wedge Q)$	$\neg P \wedge \neg Q$
F	F	T	T
F	T	T	F
T	F	T	F
T	T	F	F

The columns disagree, for example when $P = F$ and $Q = T$. Therefore the formulas are not logically equivalent. De Morgan's law has a disjunction on the right: $\neg(P \wedge Q) \equiv \neg P \vee \neg Q$. ■